



WP5.A2 Output 1 – Confirmed App Content Architecture

1. Purpose of the Content Architecture

The EcoUnity app content architecture defines how all educational, interactive, and support materials will be organised inside the mobile application.

The architecture ensures that the app is:

- easy for students aged 10–14 to use,
- practical for teachers,
- aligned with the 17 Sustainable Development Goals,
- inclusive for diverse learners,
- multilingual and accessible,
- compatible with interactive comics, micro-learning resources, quizzes, challenges, and gamification.

The app will act as the central digital access point for EcoUnity's learning materials, connecting the curriculum, branching scenario comics, and practical sustainability activities into one coherent learning journey.

2. Core Content Model

The app will be structured around **17 SDG Learning Modules**.

Each module represents one Sustainable Development Goal and follows the same internal learning structure.

Standard SDG Module Structure

Each SDG module includes:

1. SDG Introduction

2. **Learning Objectives**
3. **Interactive Comic / Scenario**
4. **Three Micro-Learning Resources**
5. **Quiz or Reflection Activity**
6. **Practical Action Challenge**
7. **Progress Tracking**
8. **Badge or Completion Reward**
9. **Teacher Notes**

This gives the app a consistent and scalable structure across all 17 topics.

3. Overall Content Hierarchy

The recommended hierarchy is:

Level 1 – App

EcoUnity mobile application.

Level 2 – Main Content Areas

The app includes the following main content areas:

1. Home
2. Learn
3. SDG Modules
4. Interactive Comics
5. Micro-Learning Resources
6. Challenges
7. Progress and Badges
8. Teacher Resources
9. Settings and Accessibility

Level 3 – SDG Modules

Each SDG module contains a complete learning package.

Level 4 – Learning Activities

Each module includes:

- short content units,
 - comics,
 - quizzes,
 - reflection prompts,
 - practical challenges,
 - teacher guidance.
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4. SDG Module Architecture

Each of the 17 SDG modules should follow this structure:

4.1 Module Header

Includes:

- SDG number
- SDG title
- SDG icon
- short student-friendly description
- estimated completion time
- completion status

4.2 Module Introduction

A short introduction explaining:

- what the SDG means,
- why it matters,
- how it connects to everyday life,
- what students can do.

4.3 Learning Objectives

Each module should define 2–4 simple learning objectives.

Example:

By the end of this module, learners will be able to:

- explain what responsible consumption means,
- identify examples of waste in school life,
- suggest one action to reduce waste,
- work with classmates to create a shared solution.

4.4 Interactive Comic

Each module links to one interactive branching scenario comic.

The comic should introduce:

- a real-life problem,
- diverse characters,
- a decision-making situation,
- social inclusion elements,
- sustainability choices.

4.5 Three Micro-Learning Resources

Each module includes three short learning resources:

MLR 1 – Understand

Introduces the topic.

MLR 2 – Explore

Shows real-life examples or stories.

MLR 3 – Act

Connects learning to action.

4.6 Quiz or Reflection

Each module includes a short activity to check understanding and encourage reflection.

This may include:

- multiple-choice questions,
- true/false questions,
- scenario-based questions,
- open reflection prompts.

4.7 Practical Challenge

Each module ends with a real-world challenge.

The challenge should encourage students to:

- collaborate,
- observe their environment,
- take practical action,
- reflect on impact.

4.8 Badge and Completion

Each module includes a completion badge.

Students earn the badge when they complete:

- the comic,
 - the three MLRs,
 - the quiz/reflection,
 - the challenge.
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5. Content Flow

The preferred student journey is:

1. Choose a Module

The learner selects an SDG topic from the module list.

2. Discover the Topic

The learner reads a short introduction and sees what they will learn.

3. Enter the Story

The learner starts the related interactive comic.

4. Learn in Small Steps

The learner completes three micro-learning resources.

5. Reflect

The learner completes a short quiz or reflection task.

6. Take Action

The learner completes a practical challenge individually or in a group.

7. Earn Progress

The learner receives progress feedback and a badge.

6. App Content Types

The app should include the following core content types:

6.1 SDG Module

The main container for each topic.

6.2 Micro-Learning Resource

A short educational content item.

Possible formats:

- text card,
- video,
- infographic,
- interactive activity,

- mini-quiz,
- audio,
- downloadable worksheet.

6.3 Interactive Comic

A branching scenario-based learning item.

6.4 Quiz

A short knowledge-check activity.

6.5 Reflection Activity

A prompt encouraging students to think about values, choices, and behaviour.

6.6 Practical Challenge

A real-world action activity.

6.7 Badge

A visual reward connected to completion and motivation.

6.8 Teacher Note

A short guidance item helping teachers use content in class.

6.9 Accessibility Note

A content-specific note identifying adaptations or support needs.

7. Metadata Structure

Each content item should include metadata to support organisation, translation, filtering, and reporting.

Recommended metadata fields:

- content ID,
- title,

- content type,
 - SDG number,
 - SDG title,
 - language,
 - age group,
 - estimated duration,
 - difficulty level,
 - learning objective,
 - inclusion theme,
 - sustainability theme,
 - partner responsible,
 - content status,
 - translation status,
 - accessibility status,
 - last updated date.
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8. Multilingual Architecture

The app should support multilingual content from the beginning.

Each content item should have:

- one master version,
- translated language versions,
- language-specific title,
- language-specific body text,
- language-specific media where needed,
- translation review status.

Recommended language structure:

- English as master content language,

- Polish,
 - German,
 - Finnish,
 - Romanian,
 - Spanish,
 - Ukrainian where relevant.
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9. Inclusion Architecture

Inclusion should not be treated as a separate section only. It should be embedded into all content.

Each module should include:

- diverse names and characters,
- cooperation prompts,
- intercultural reflection,
- accessible language,
- practical low-cost activities,
- group work options,
- alternatives for students with different learning needs.

Example inclusion prompt:

Work with someone you do not usually work with. Listen to their idea first, then agree on one shared action.

10. Gamification Architecture

Gamification should support motivation without creating pressure or unhealthy competition.

Recommended elements:

- SDG module badges,
- progress bars,

- challenge completion marks,
- final EcoUnity badge,
- encouraging feedback messages.

Avoid:

- public ranking,
 - competitive leaderboards,
 - performance comparison between students.
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11. Teacher Support Architecture

Teacher support should be included as a lightweight but useful layer.

Each SDG module should include teacher guidance with:

- classroom use suggestions,
- discussion questions,
- group work tips,
- inclusion adaptations,
- estimated duration,
- connection to curriculum,
- suggested follow-up activity.

Teacher resources should be accessible from:

- each SDG module,
 - a separate Teacher Resources screen.
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12. Recommended Module Count and Content Volume

The confirmed full content architecture should include:

17 SDG modules

One module per SDG.

51 micro-learning resources

Three micro-learning resources per SDG module.

17 interactive comic links

One comic or scenario per SDG module.

17 quizzes or reflections

One per SDG module.

17 practical action challenges

One per SDG module.

17 SDG badges

One completion badge per SDG module.

1 final EcoUnity badge

Awarded after completing a defined number of modules or all modules.

13. Minimum Prototype Architecture

For the first app prototype, the minimum content architecture should include:

1. Home screen
2. SDG module list
3. One complete SDG module
4. Three micro-learning resources
5. One comic link
6. One quiz or reflection
7. One challenge
8. One badge
9. One teacher note
10. Language selection

11. Basic progress tracking

This will allow the consortium to test the full learning flow before producing all 17 modules.

14. Recommended First Prototype Module

The first complete prototype module should be:

SDG 12 – Responsible Consumption and Production

Reason:

This topic is easy for students aged 10–14 to understand and can be directly connected to everyday school life.

Suggested prototype package:

- Module title: Responsible Consumption and Production
 - Comic: The Zero-Waste School Day
 - MLR 1: What is waste?
 - MLR 2: How can schools reduce waste?
 - MLR 3: What can I do this week?
 - Quiz: 5 short questions
 - Challenge: Create a zero-waste classroom plan
 - Badge: Responsible Consumer Badge
 - Teacher note: Running the challenge in mixed groups
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15. Confirmed Architecture Summary

The EcoUnity app content architecture is confirmed as a modular SDG-based learning system.

Each SDG module combines:

- story-based learning,
- micro-learning,
- reflection,

- practical action,
- inclusion prompts,
- teacher support,
- gamified progress.

This structure ensures that the app is educationally meaningful, technically manageable, scalable, multilingual, inclusive, and aligned with the project's goals of sustainability education, social mixing, and active citizenship.